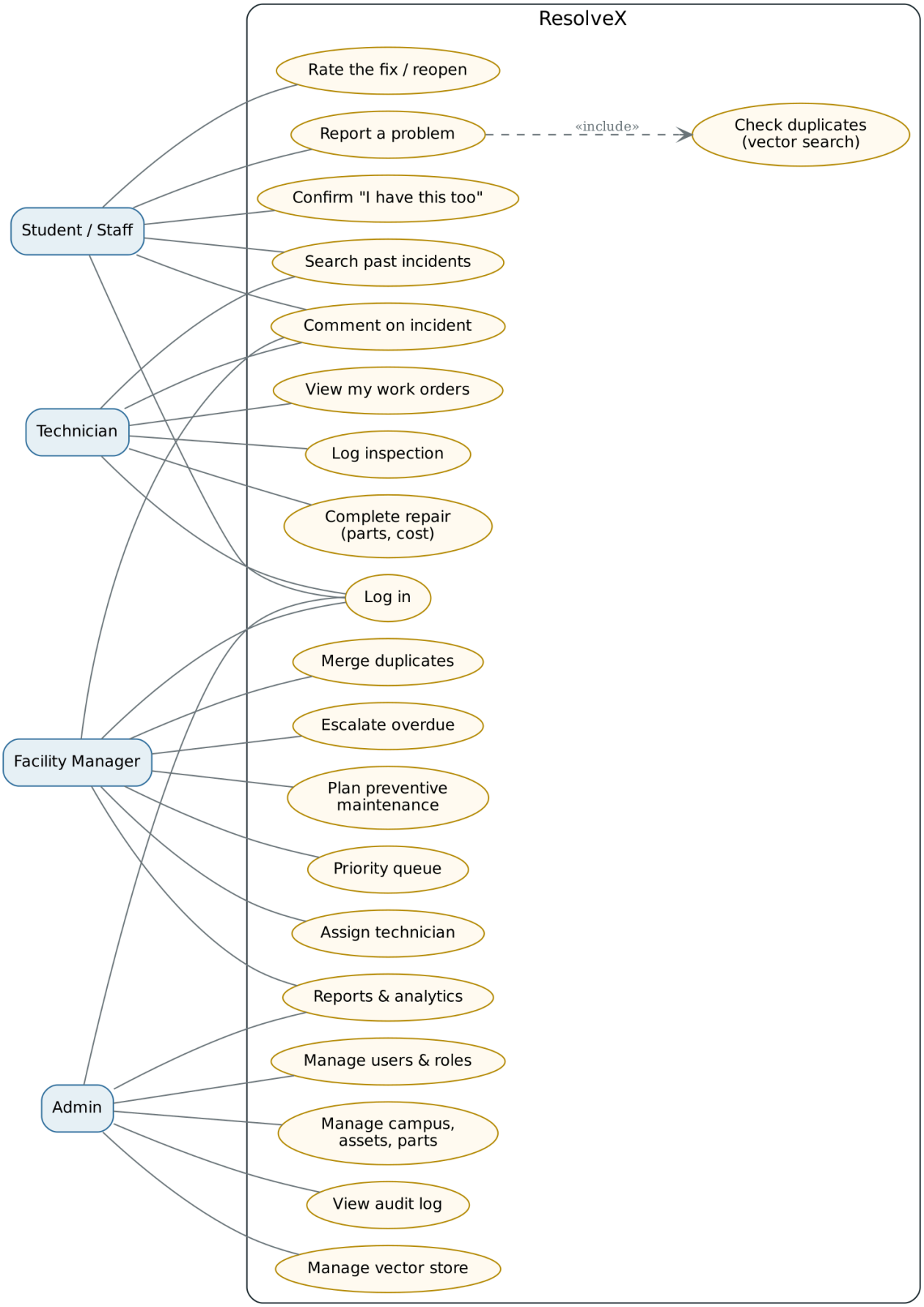
ResolveX Database Statistics Report

ResolveX recorded 171 incidents between May and October 2026; 88% were resolved, 82.7% within their SLA deadline, at a total maintenance cost of ₹4,65,420. All figures below are computed live from the PostgreSQL and MongoDB databases (demo dataset loaded by `npm run db:setup`).

Use cases

Four actors use ResolveX. Each use case below is backed by a database object that does the work.


ResolveX
Rate the fix / reopen
Report a problem
«include»
Check duplicates (vector search)
Confirm "I have this too"
Search past incidents
Comment on incident
View my work orders
Log inspection
Complete repair (parts, cost)
Log in
Merge duplicates
Escalate overdue
Plan preventive maintenance
Priority queue
Assign technician
Reports & analytics
Manage users & roles
Manage campus, assets, parts
View audit log
Manage vector store
Student / Staff
Technician
Facility Manager
Admin

Actor	Use case	What happens in the database
Student / Staff	Report a problem	INSERT into incidents + incident_embeddings in one transaction; trigger computes priority and deadline
Student / Staff	Check duplicates (included in reporting)	fn_detect_duplicates compares the text by meaning (pgvector)
Student / Staff	Confirm “I have this problem too”	INSERT incident_reporters; trigger raises affected users and priority
Student / Staff	Rate the fix / reopen	UPDATE status; state-machine trigger allows only valid changes
Technician	Log inspection	insertOne into MongoDB incident_notes, checked by the category’s validator
Technician	Complete repair	CALL sp_complete_work_order: stock, cost, status in one transaction
Facility Manager	Priority queue	SELECT from v_incident_details ordered by priority score
Facility Manager	Assign technician	CALL sp_assign_technician (best skilled, least busy, row locks)
Facility Manager	Merge duplicates / escalate overdue	CALL sp_mark_duplicate (recursive CTE) / sp_escalate_overdue
Facility Manager	Reports and analytics	8 SQL reports, 7 MongoDB pipelines, views and a materialized view
Admin	Manage users, campus, assets, parts	INSERT / UPDATE with CHECK, UNIQUE and FK validation; audit trigger records changes
Admin	View audit log / manage vector store	audit_log (RLS: admin only); DELETE incident_embeddings (RLS: admin only)
All	Log in	fn_login_lookup reads the bcrypt hash; Row-Level Security then limits every query to the user’s role

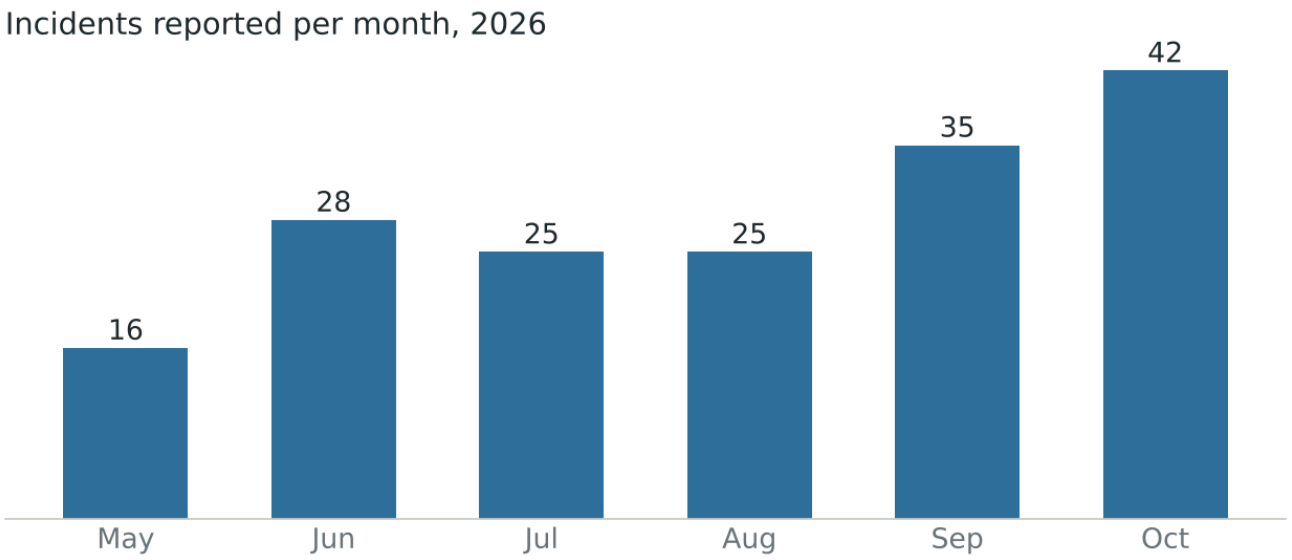
Database at a glance

PostgreSQL holds 20 tables and MongoDB holds 3 collections. The largest tables are the history and vector tables, because every incident gets a status trail and an embedding.

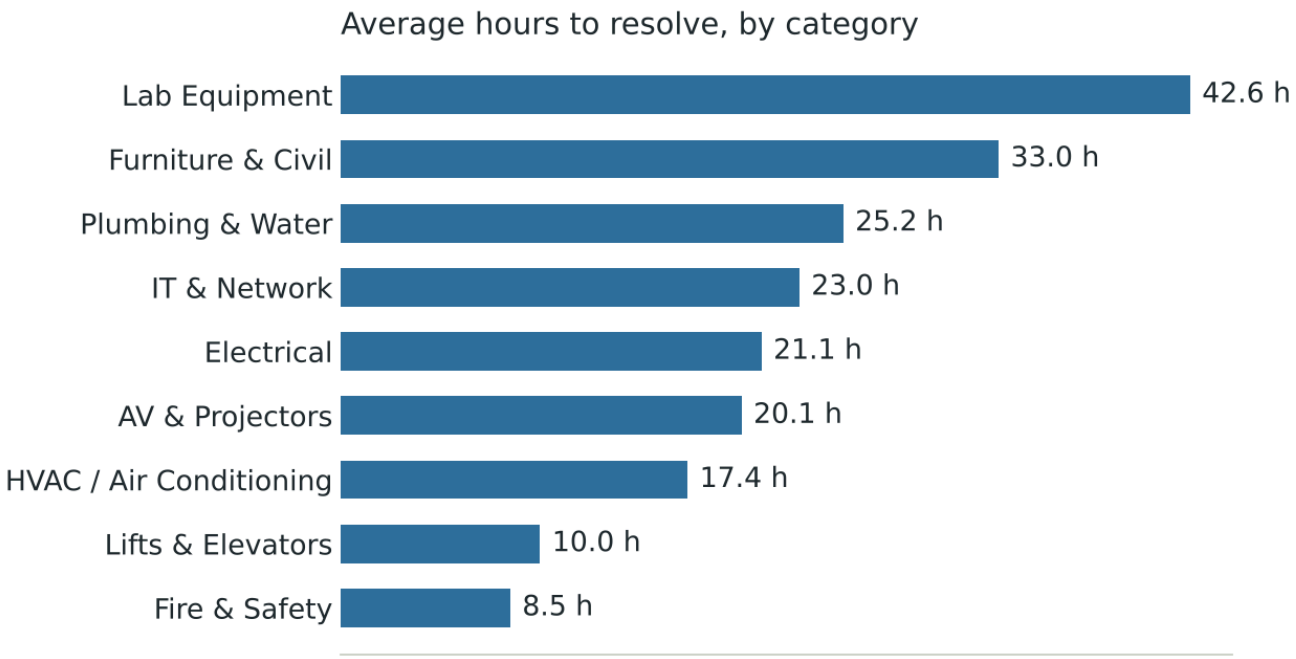
Measure	Value
Incidents (May-Oct 2026)	171
Status-history rows	773
Vector embeddings (384 numbers each)	171
Work orders	155
Maintenance records	140 (132 corrective, 8 preventive)
Users	21 (11 students/staff, 7 technicians, 2 managers, 1 admin)
Buildings / rooms / assets	8 / 30 / 35
Spare parts	24 types
MongoDB documents	345 notes (192 comments, 153 inspection logs), 9 templates, 260 search events

Incidents

Reports grew from 16 in May to 42 in October, as more of the campus started using the system. On average an incident is resolved in 21.2 hours, and 82.7% are resolved within their SLA deadline.



By status: 134 closed, 16 resolved (waiting for the reporter to confirm), 9 merged as duplicates, and 12 still active (7 open, 2 assigned, 2 in progress, 1 on hold).



Category	Incidents	SLA met
HVAC / Air Conditioning	40	80.6%
IT & Network	35	80.6%
Plumbing & Water	29	80.8%
AV & Projectors	24	85.7%
Electrical	21	88.9%

Category	Incidents	SLA met
Lifts & Elevators	6	100%
Fire & Safety	6	66.7%
Lab Equipment	6	80.0%
Furniture & Civil	4	100%

HVAC and IT & Network produce 44% of all incidents. Lab equipment takes longest to fix (42.6 h), because parts often have to be ordered.

By priority (computed by the priority function and trigger):

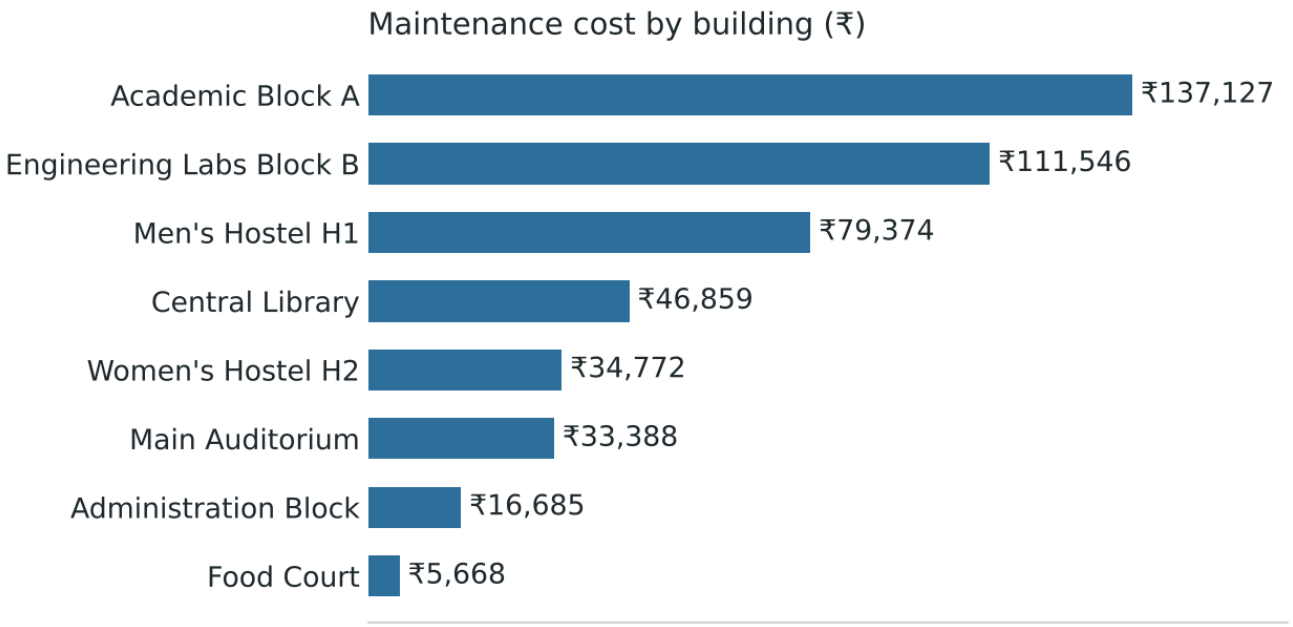
Priority	Incidents	Average score	SLA met
P1 (fix within 8 h)	20	83.8	76.5%
P2 (24 h)	92	70.5	80.2%
P3 (72 h)	46	54.5	88.9%
P4 (168 h)	4	40.6	100%

P1 incidents have the tightest deadline and the lowest SLA rate. They are the first place to improve, and the reason the system escalates overdue incidents automatically.

Reporter satisfaction: 120 ratings, average 4.18 out of 5 (52 five-star, 43 four-star, 20 three-star, 5 two-star).

Maintenance cost, spare parts and technicians

Repairs cost ₹4,65,420 in total: ₹3,10,500 in parts (67%) and ₹1,54,920 in labour, an average of ₹3,324 per job. Academic Block A and the Engineering Labs account for more than half the spend.



Most expensive parts used:

Part	Quantity used	Value
Wi-Fi Access Point	6	₹87,000
UPS Battery 12V 42Ah	10	₹55,000
Projector Lamp	4	₹27,200
SFP Transceiver Module	5	₹21,000
RO Membrane 75 GPD	11	₹17,600

At or below the reorder level (low-stock trigger has alerted the managers): AC Indoor Fan Motor (2 left), Projector Lamp (2), AC Run Capacitor 45 uF (3, reorder level 5), Geyser Heating Element (3). Reordering all four costs about ₹25,850.

Technician performance:

Technician	Jobs completed	Avg turnaround	Avg reporter rating
Lakshmi Prasad	38	16.9 h	4.00
Deepak Joshi	29	23.6 h	4.33
Priya Shetty	25	16.0 h	4.29
Ravi Kumar	22	14.2 h	4.47
Imran Khan	20	26.0 h	4.24

Technician	Jobs completed	Avg turnaround	Avg reporter rating
Suresh Naidu	10	25.8 h	3.78
Manjunath Gowda	6	6.9 h	3.67

Ravi Kumar combines a fast turnaround with the highest rating. Lakshmi Prasad carries the largest workload.

Asset health and predictive maintenance

Of 35 assets, 5 are high risk, 21 medium and 9 low. The risk score (function `fn_asset_risk_score`) combines past failures, overdue maintenance, age and criticality. The mean time between failures (MTBF) comes from the `LAG()` window function in the view `v_asset_health`.

Asset	Building	Failures	MTBF	Predicted next failure	Risk
Ducted AC, Library Reading Hall	Central Library	7	17.4 days	1 Oct 2026 (overdue)	79.2 High
Central AC Plant, Auditorium	Main Auditorium	7	23.5 days	21 Oct 2026 (overdue)	78.9 High
Fire Alarm Panel, Labs Block	Engineering Labs Block B	4	47.1 days	26 Nov 2026	78.7 High
Passenger Lift, Academic Block A	Academic Block A	2	73.1 days	12 Jan 2027	76.0 High
Split AC 2 Ton, Computer Lab 1	Engineering Labs Block B	13	13.5 days	14 Nov 2026	75.1 High

The Computer Lab 1 AC failed 13 times, roughly every two weeks. Replacing it is likely cheaper than continuing to repair it. The procedure `sp_schedule_preventive_maintenance` creates inspection jobs for every high-risk asset.

MongoDB statistics

These come from MongoDB aggregation pipelines on the `incident_notes` and `search_events` collections.

Inspection verdicts (153 logs): 107 needed parts (70%), 37 were fixed on site (24%), 4 found no fault, 3 needed an outside vendor and 2 are being monitored. Seven in ten visits need a part, so parts stock directly decides how fast repairs finish.

Most failed inspection checks:

Equipment	Check that failed	Times
AV & Projectors	Cables and adapters OK	17
HVAC	Filter clean	12
IT & Network	Firmware up to date	9
IT & Network	Device powered (PoE)	9

Search log (260 searches): 161 hybrid, 67 meaning-only and 32 keyword searches. The most searched are “ac not cooling” (30), “wifi slow” (28) and “room is too hot” (21). Searches that found nothing were mostly outside the system’s scope, such as “hostel mess timing” (5) and “canteen food quality” (4), which suggests adding a food-service category.

Key findings

1. **Demand is rising:** monthly reports grew 2.6× from May (16) to October (42).
2. **HVAC and network are the hot spots:** 75 of 171 incidents (44%).
3. **P1 deadlines are missed most often** (76.5% met vs 82.7% overall); automatic escalation targets exactly these.
4. **Parts drive cost and delay:** 67% of the money and 70% of inspection verdicts involve a part; 4 parts are already at the reorder level.
5. **Five assets are high risk;** the Computer Lab 1 AC (13 failures) is a replacement candidate.
6. **Users are satisfied:** average rating 4.18 out of 5.

The SQL behind the numbers

Every table above comes from a query like this one (average resolution time and SLA compliance per category; it is report 1 on the app’s Reports page):

```
SELECT c.category_name,
       COUNT(*) AS incidents,
       ROUND(AVG(EXTRACT(EPOCH FROM (i.resolved_at - i.reported_at)) / 3600)
             FILTER (WHERE i.resolved_at IS NOT NULL), 1) AS avg_hours,
       ROUND(100.0 * COUNT(*) FILTER (WHERE i.resolved_at <= i.sla_due_at)
             / NULLIF(COUNT(*) FILTER (WHERE i.resolved_at IS NOT NULL), 0),
1) AS sla_pct
  FROM incidents i
  JOIN asset_categories c ON c.category_id = i.category_id
 GROUP BY c.category_name
 ORDER BY incidents DESC;
```

Other sources: `v_asset_health` (risk, MTBF), `v_low_stock_parts`, `maintenance_records` joined to `assets`, `rooms` and `buildings` (cost by building), and MongoDB `$group` / `$unwind` pipelines (verdicts, failed checks, search terms).